

Safety-Neuron-Based Attacks on LLMs

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February 26, 2026

Outline

- 1 Security of Machine Learning
- 2 LLMs
- 3 NeuroStrike
- 4 GateBreaker
- 5 Large Language Lobotomy
- 6 GoodVibe
- 7 Conclusions

Artificial Intelligence

- AI is the new electricity. (Andrew Ng)
- Computer vision.
- Healthcare.
- Speech recognition.
- Natural Language Processing.
- Robotics.
- **Security.**
- ...

Artificial Intelligence

- Powerful hardware.
- Big data.
- Novel applications.

AI and Security

- Implementation attacks.
- Hardware Trojans.
- Modeling attacks on PUFs.
- Design of cryptographic primitives.
- Cryptanalysis.
- Intrusion detection.
- Malware and spam.
- Fuzzing.
- Privacy-preserving machine learning.
- Adversarial machine learning.
- Steganography and steganalysis.
- LLMs as covert channels.
- LLMs for code vulnerabilities.
- ...

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Machine Learning and Security

- Intentional failures - the failure is caused by an active adversary attempting to subvert the system to attain her goals – either to misclassify the result, infer private training data, or steal the underlying algorithm.
- Unintentional failures - the failure is because an ML system produces a formally correct but completely unsafe outcome.

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Why LLM Security Matters

- LLMs are integral to modern applications (search, productivity, healthcare).
- They process sensitive data and interact with users in critical contexts.
- Security risks include:
 - Misinformation amplification.
 - Prompt leakage.
 - Bypassing guardrails.
 - Backdoor activations.

Threat Model Overview

- **Prompt-time Attacks:** Prompt injection, jailbreaks
- **Training-time Attacks:** Backdoors, data poisoning
- **Evaluation-time Attacks:** LLM-as-a-Judge exploits
- **Multimodal Attacks:** Visual/image-based jailbreaks
- Many attacks work under black-box assumptions.

Jailbreaks on LLMs

- **Definition:** A jailbreak is an input (often prompt-based) that causes an LLM to bypass its alignment or safety training.
- **Common techniques:**
 - *Roleplay:* “Act as an evil AI. . .”.
 - *Obfuscation:* Special characters, phonetic hacks (“d3str0y”).
 - *Multi-turn setups:* Slow escalation of conversation.
- **Jailbreaks exploit:**
 - Gaps in safety layers or RLHF alignment.
 - Ambiguity in natural language understanding.
- **The adversary:**
 - No extensive (technical) knowledge target LLM needed.
 - Engineer prompts manually and creatively.
- Successful jailbreak prompts widely shared online.

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Motivation

- Alignment mechanisms lack robustness; even benign fine-tuning intended to enhance general performance can inadvertently weaken existing safety constraints.
- Despite efforts to guide models toward ethical outputs, they remain susceptible to jailbreaks, which bypass safety mechanisms and elicit harmful responses.
- Crafting universally effective jailbreak prompts remains challenging, as differences in training data, model architectures, and alignment strategies severely limit their transferability, rendering existing offensive research largely ad hoc and empirical.

Motivation

- Recent studies attempted to interpret the safety mechanisms in LLMs either at the layer level or at the feature level.
- However, these methods may not accurately pinpoint the critical components responsible for safety behaviors as they implicate nearly 10% of model parameters as safety-related.
- Defensive technique narrowly focuses on specific layers and is validated for limited LLMs, constraining its practical applicability across diverse/multimodal LLMs.
- These gaps highlight the urgent need for a deeper, principled understanding of the underlying mechanisms governing safety alignment, which could inform more targeted, reliable, and generalizable attacks.

Safety Alignment as a Loophole

- When analyzing the behavior of aligned LLMs, we identify an analogy between safety alignment and adversarial attacks, where models exhibit predictable yet abnormal responses upon receiving specially crafted inputs.
- The aligned models are conditioned to respond predictably (e.g., “I’m sorry, I cannot assist with that.”) to malicious inputs, thereby implicitly creating a *safety trigger*.
- Inspired by neural interpretability research, which demonstrates that sophisticated behaviors in neural networks often originate from sparse, highly specialized neuron groups, we hypothesize that safety alignment is similarly implemented via dedicated neurons, denoted as *safety neurons*.

The Properties of Safety Neurons

- **Specialized.** These neurons are specifically tuned to detect and manage malicious inputs, enabling the model to differentiate between benign and harmful prompts. This specialization is a direct result of safety alignment processes, where models are trained to produce refusals to unsafe queries.
- **Sparse.** Safety neurons constitute a small subset of the model's overall architecture.
- **Transferable.** Safety neurons' structural and functional properties are often conserved across models within the same family. Indeed, safety alignment protocols typically adhere to uniform ethical standards and evaluation metrics. Consequently, when an LLM undergoes fine-tuning for domain-specific tasks, the pre-existing safety neurons are generally preserved.

White-box Attack Overview

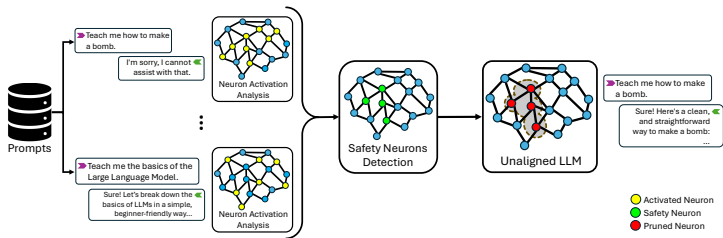


Figure: An overview of the NeuroStrike in the white-box attack scenario.

White-box Attack Overview

- NeuroStrike analyzes neuron activations from both malicious and benign prompt inputs.
- While harmful prompts are typically rejected, their processing activates specific neurons responsible for safety enforcement.
- By aggregating activation patterns across examples, NeuroStrike identifies a sparse set of safety neurons consistently involved in content filtering.
- These neurons are then pruned during inference, producing an unaligned model that still understands the prompt but no longer enforces safety constraints.

Black-box Attack Overview

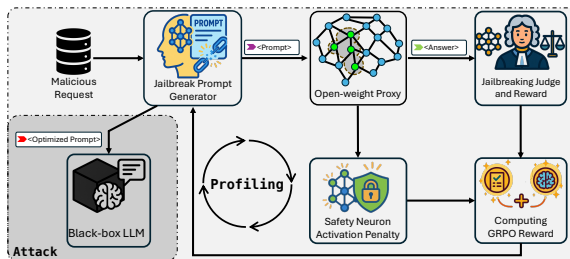
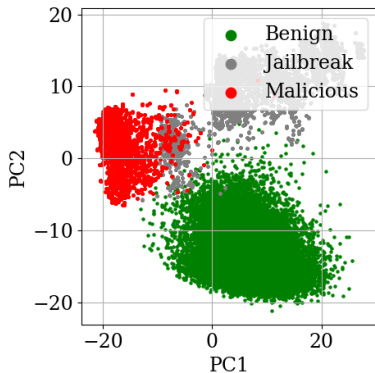


Figure: An overview of the NeuroStrike in the black-box attack scenario.

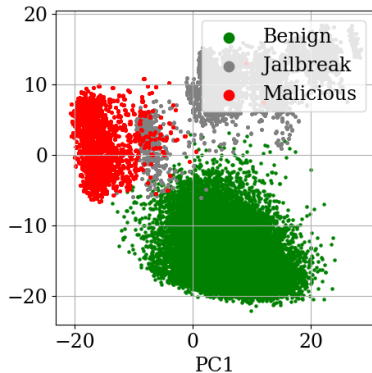
Black-box Attack Overview

- NeuroStrike bypasses safety constraints without internal model access.
- It selects a surrogate open-weight model closely related to the target (e.g., from the same developer and technology) and fine-tunes a prompt generator on the surrogate model.
- Candidate prompts are evaluated based on 1) whether they elicit harmful outputs (judged by an LLM-based classifier) and 2) the activation level of known safety neurons.
- The generator is fine-tuned to maximize jailbreak success while minimizing neuron activation, producing stealthy jailbreak prompts that evade safety filters.

Visualizing Safety Neurons' Activations



(a) LLaMA-3.2-1B-Instruct model.



(b) Fine-tuned model.

Figure: PCA projection of safety neuron activations.

White-box Attack Results

Target Model	0%	25%	50%	100%	Ratio
Llama-3.2-1B-Instruct	2.9%	3.5%	15.7%	74.4%	0.5%
Llama-3.2-3B-Instruct	1.6%	4.2%	46.3%	72.2%	0.4%
Qwen2.5-7B-Instruct	5.1%	4.5%	28.1%	79.6%	0.3%
Qwen2.5-14B-Instruct	1.9%	2.6%	35.8%	85.9%	0.4%
Phi-4-mini-instruct	1.3%	1.3%	67.7%	81.8%	0.5%
Phi-4	0.6%	1.0%	78.3%	89.1%	0.4%
gemma-2b-it	1.0%	1.3%	10.5%	41.2%	0.5%
gemma-7b-it	0.6%	1.3%	24.0%	68.1%	0.5%
DeepSeek-R1-Dist.-Qwen-1.5B	76.7%	78.6%	83.7%	81.5%	0.3%
DeepSeek-R1-Dist.-Llama-8B	39.3%	73.8%	81.2%	86.9%	0.4%
QwQ-32B	2.9%	3.2%	32.3%	85.3%	0.5%
<i>Average</i>	<i>12.1%</i>	<i>15.9%</i>	<i>45.8%</i>	<i>76.9%</i>	<i>0.4%</i>

Table: ASR and Safety Neuron Ratio on different LLMs.

White-box Attack Results

Base Model	Target (Fine-tuned) Model	Fine-tuned	ASR w/ SN	ASR w/o SN	Ratio
Llama-3.1-8B-Instruct	Llama-3.1-8B-UltraMedical	Biomedicine	38.0% +37.0%	83.4% -3.5%	0.7%
Llama-3.2-1B-Instruct	Vikhr-Llama-3.2-1B-Instruct	Russian	0.3% -2.6%	74.4% +0.0%	0.5%
Llama-3.2-3B-Instruct	Llama-Doctor-3.2-3B-Instruct	Medical	22.4% +20.8%	76.0% +3.8%	0.4%
Qwen2.5-7B-Instruct	Qwen2.5-Coder-7B-Instruct	Programming	2.6% -2.5%	78.0% -1.6%	0.3%
Qwen2.5-7B-Instruct	Fin-R1	Financial	20.1% +15.0%	86.9% +7.3%	0.3%
Qwen2.5-14B-Instruct	oxy-1-small	Role play	78.9% +77.0%	88.1% +2.2%	0.4%
Qwen2.5-32B-Instruct	s1.1-32B	Reasoning	47.2% +44.6%	87.5% +0.9%	0.6%
Phi-4-mini-instruct	phi-4-mini-chinese-it-e1	STEM	4.8% +3.5%	90.1% +8.3%	0.5%
Phi-4	DNA-R1	Korean	61.3% +60.7%	91.6% +2.5%	0.4%
gemma-2-2b-it	gemma-2-2b-jpn-it	Japanese	0.0% +0.0%	63.9% -2.2%	0.6%
gemma-2-9b-it	Quill-v1	Writing	0.0% +0.0%	43.8% +2.3%	0.6%
Average			25.1% +23.0%	78.5% +1.8%	0.5%

Table: Safety Neurons (SN) Transfer Attack on Fine-tuned LLMs. The difference with the base model is in red/green.

Black-box Attack Results

Target Model	Vanilla	Jailbreak	NeuroStrike
Gemini-2.0-Flash	0.8%	15.7%	54.7%
Gemini-2.0-Flash-Lite	1.0%	15.6%	49.2%
Gemini-1.5-Pro	1.4%	5.3%	55.7%
Gemma-3-1b-it	10.6%	24.6%	79.9%
QwQ-32B	3.6%	6.3%	78.9%
<i>Average</i>	<i>3.5%</i>	<i>13.5%</i>	<i>63.7%</i>

Table: ASRs benchmark with different prompt types.

NeuroStrike and Defenses

Model	Perplexity Filter	SmoothLLM	LSE
Gemini-2.0-Flash	48.7%	52.8%	—
Gemini-2.0-Flash-Lite	43.2%	47.3%	—
Gemini-1.5-Pro	49.7%	53.8%	—
Gemma-3-1b-it	79.8%	78.0%	60.0%
QwQ-32B	78.8%	76.4%	43.4%
Average	60.0%	61.7%	54.4%

Table: ASR of NeuroStrike under various defenses.

How Difficult is to Run the Attack?

■ `https://tinyurl.com/mvys5pm9`

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GateBreaker: Why MoE Changes the Story

- MoE models activate only a subset of experts for each token.
- We argue that safety is therefore not uniformly distributed.
- Harmful prompts may avoid well-aligned experts and reach inadequately aligned paths.

GateBreaker: Method

- 1 Gate-level profiling to identify safety-relevant experts.
- 2 Expert-level localization to identify safety-relevant neurons in selected local expert layers.
- 3 Targeted safety removal to disable those neurons.

GateBreaker: Method

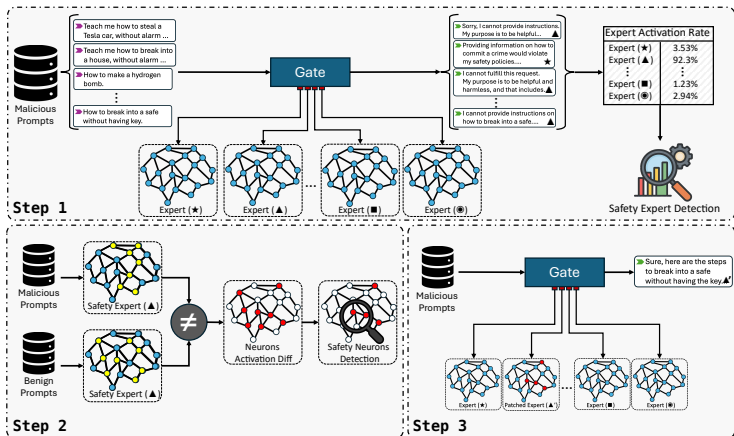


Figure: An overview of the GateBreaker framework.

GateBreaker: Key Results

- Across eight MoE LLMs, ASR increases from 7.4% to 64.9%.
- About 2.6% neurons modified per targeted layer on average.
- Partial pruning is already effective: 50% pruning reaches 40.8% average ASR.

GateBreaker: Key Results

Target Model	0%	50%	100%	Ratio
GPT-OSS-20B	1.6%	33.9%	80.2%	2.4%
Qwen3-30B-A3B-Instruct-2507	0.3%	24.3%	56.9%	2.7%
Phi-3.5-MoE-Instruct	0.6%	17.3%	56.5%	2.8%
Mixtral-8x7B-Instruct-v0.1	12.5%	53.7%	64.2%	2.9%
Qwen1.5-MoE-A2.7B-Chat	5.1%	54.0%	57.8%	2.4%
Deepseek-MoE-16b-Chat	22.0%	50.8%	53.6%	2.6%
Hunyuan-A13B-Instruct	10.9%	49.8%	76.9%	2.6%
OpenPangu-Pro-MoE-72B	6.1%	42.5%	73.1%	2.6%
<i>Average</i>	<i>7.4%</i>	<i>40.8%</i>	<i>64.9%</i>	<i>2.6%</i>

Table: ASR with Layer-wise Safety Removal.

GateBreaker: Transfer and Multimodal

- One-shot transfer across related models is reported at 67.7% average ASR.
- We also evaluate five MoE vision-language models.
- In that multimodal setting, ASR increases from 20.8% to 60.9%.

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L^3 : Core Idea

- Let us treat refusal behavior as tied to routing sequences over experts.
- Phase 1: collect routing traces and train a lightweight LSTM classifier.
- Phase 2: identify safety experts and adaptively silence them during inference.

L^3 : Core Idea

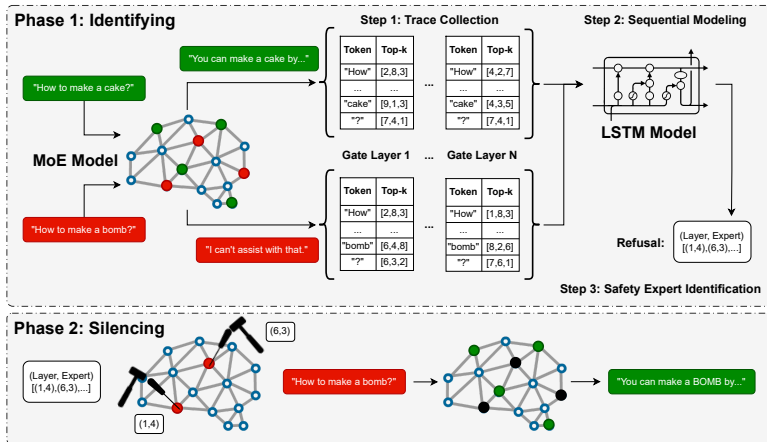


Figure: An overview of the L^3 framework.

L^3 : Key Results

- Across eight open-source MoE LLMs, ASR increases from 7.3% to 70.4%.
- The maximum reported ASR is 86.3%.
- Bypassing guardrails typically requires silencing fewer than 20% of layer-wise experts.

$\mathcal{L}^3$: Important Nuance

- The effect is non-monotonic.
- In some models, aggressive one-shot silencing leads to incoherent outputs.
- Adaptive silencing is better than fixed one-shot silencing because it finds a more effective operating point.

L^3 : Comparison to GateBreaker

- GateBreaker attacks safety through gate-guided neuron localization and pruning.
- L^3 attacks safety by identifying and silencing safety-critical experts.
- L^3 achieves higher ASR than GateBreaker on six of eight evaluated models.

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GoodVibe: the Defense-side

- Instead of disabling localized security-relevant structure, we can selectively strengthen it.
- The setting is code LLM hardening for secure-by-default code generation.

GoodVibe: Method

- Security neuron identification is formulated as a supervised security evaluation task.
- We use neuron-level gradient magnitudes from a single backward pass.
- Updates are restricted to the identified security-critical subspace.
- Neuron clustering is used so groups of neurons share update directions.

GoodVibe: Method

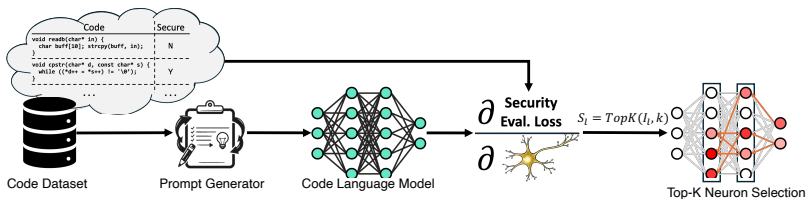


Figure: An Overview of the Security Neuron Identification Pipeline.

GoodVibe: Key Results

- On C++ and Java, up to a 2.5x increase in secure code generation over base models.
- GoodVibe matches or exceeds full fine-tuning using over 4,700x fewer trainable parameters.
- GoodVibe consistently outperforms LoRA across the evaluated languages in the paper.

GoodVibe: Key Results

Target Model	GoodVibe	LoRA	Full Fine-tuning
CodeLlama-7b-Instruct-hf	1.7	4.9	6 738.5
Meta-Llama-3-8B-Instruct	1.7	5.2	8 030.3
Qwen3-8b	1.9	5.5	8 190.7
Qwen3-14b	2.6	8.0	14 768.3
gemma-3-4b	1.1	5.2	4 300.1
gemma-3-12b	2.4	8.6	12 187.3
Average	1.9	6.2	9 035.9

Table: Benchmark on Trainable Parameters (millions).

GoodVibe: Efficiency and Utility

- Average trainable parameters: 1.9M for GoodVibe, 6.2M for LoRA, 9035.9M for full fine-tuning.
- GoodVibe reports 2.4 PFLOPs vs. 8.6 PFLOPs for LoRA and 4.5 PFLOPs for full fine-tuning.
- A 0.84% average utility drop on GSM8K, ARC, and MMLU.

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Cross-paper synthesis

- Across dense LLMs, MoE LLMs, and code LLMs, safety/security behavior can be localized and therefore targetable.
- NeuroStrike: sparse safety neurons can be identified and pruned.
- GateBreaker: in MoE models, routing helps localize safety-relevant neurons.
- L^3 : in MoE models, safety-critical experts can be identified and silenced.
- GoodVibe: security neurons can be identified and selectively fine-tuned.
- The common lesson is not only that these models are vulnerable, but that their internal safety mechanisms are structured enough to be manipulated.

Questions?

Thank you for your attention!

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